Ch 3.1: More Linear Regression

Lecture 5 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Fri, Jan 23, 2026

Announcements

Last time:

- Started 3.1 - Simple linear regression (least squares)

Announcements:

- Homework #1 Due Sun, Jan 25
- Homework #2 Due Sun, Feb 8

Covered in this lecture

- Confidence interval, hypothesis test, and p-value for coefficient estimates
- Residual standard error (RSE)
- R squared

Section 1

Last time

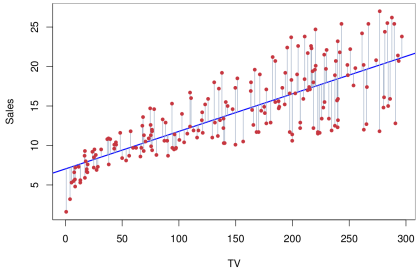
Setup

- Predict Y on a single predictor variable X

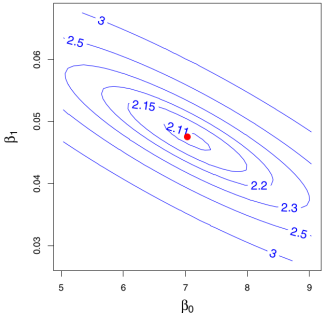
$$Y \approx \beta_0 + \beta_1 X$$

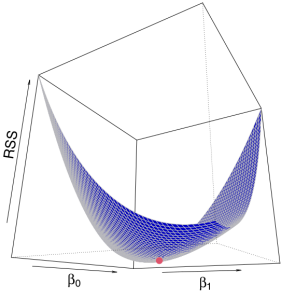
- "$\approx$" .... "is approximately modeled as"

- Given $(x_1, y_1), \cdots, (x_n, y_n)$
- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be prediction for Y on ith value of X.
- $e_i = y_i - \hat{y}_i$ is the ith residual



Least squares criterion: RSS





Residual sum of squares RSS is

$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Least squares criterion

Find β_0 and β_1 that minimize the RSS.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$
$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

Section 2

Assessing Coefficient Estimate Accuracy

Bias in estimation

Analogy with mean

Everything here is about determining if linear model is doing a good job
Explain sample vs population.

- Assume a true value μ^*
- An estimate from training data $\hat{\mu}$
- The estimate is unbiased if $E(\hat{\mu} = \mu^*)$

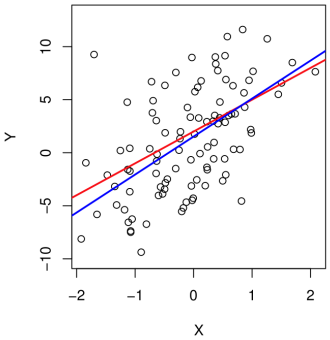
- Sample mean is unbiased for population mean:

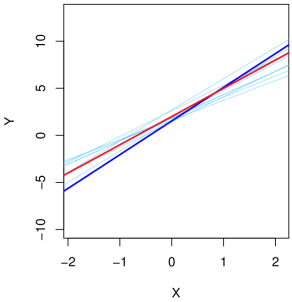
$$E(\hat{\mu}) = E\left(\frac{1}{n}\sum_i X_i\right) = \mu$$

- Standard variance estimate is biased *(under estimation)*

$$E(\hat{\sigma}^2) = E\left[\frac{1}{n}\sum_i (X_i - \overline{X})^2\right] \neq \sigma^2$$

Linear regression is unbiased





- 100 data points drawn from $Y = 2 + 3X + \varepsilon$
- ε drawn from normal distribution with mean 0
- Red line is true relationship, blue is least squares estimate
- Repeat this 10 times and plot all the found lines (in variations of blue)
- The resulting models are slightly different but are all around the red true relationship

Variance in estimation

Continuing analogy with mean

- True value μ^*
- Estimate from training data $\hat{\mu}$
- Variance of sample mean $\mathrm{Var}(\hat{\mu}) = \mathrm{SE}(\hat{\mu})^2 = \frac{\sigma^2}{n}$

- Standard error
- The more data you have, the smaller variance, the better the estimate
- the law of large numbers (LLN)

Variance of linear regression estimates

- Variance of linear regression estimates:

$$\mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^n (x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Residual standard error is an estimate of σ

$$RSE = \sqrt{RSS/(n-2)}$$

if data are more spreadout the SE is smaller

Coding group work

Run the section titled "Simulating data"

Didn't quite finish this in class Fall 2023, had them just run the part of the code that got the simulation up and running to the point of having a histogram. Moved the variance slides to next class

Confidence Interval

The 95% confidence interval for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$

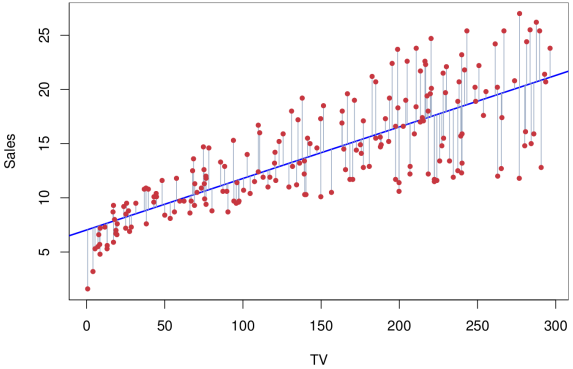
Same form works for β_0

Interpretation:
There is approximately a 95% chance that the interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain β_1 where we repeatedly approximate $\hat{\beta}_1$ using repeated samples.

CI in Advertising data



For the advertising data set, the 95% CIs are:

- β_1 :: $[0.042, 0.053]$
  - First line through [0,8] and [300,20.6]
  - Second line through [0,7] and [300,21.9]
- β_0 :: $[6.130, 7.935]$

Hypothesis testing

H_0: There is no relationship between X and Y *(null hypothesis)*

H_1: There is some relationship between X and Y *(alternative hypothesis)*

$$H_0 : \beta_1 = 0$$
$$H_A : \beta_1 \neq 0$$

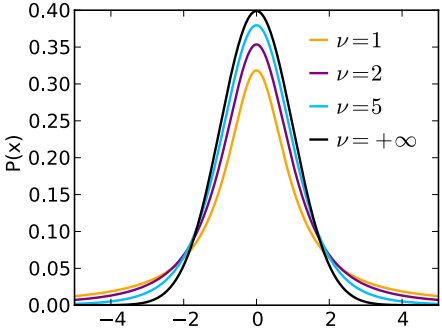
since if $\beta_1 = 0$ then the model reduces to $Y = \beta_0 + \varepsilon$, and thus X is not associated with Y

Test statistic and p-value

Test statistic:

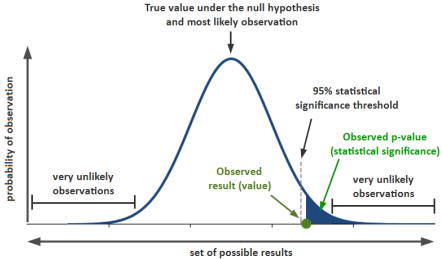
$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$$

t-distribution with $n - 2$ degrees of freedom



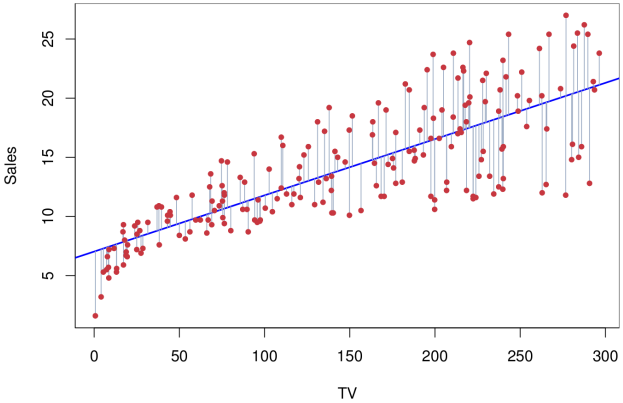
A small p-value indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance, in the absence of any real association between the predictor and the response.

Draw me:



Advertising example

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	7.0325	0.4578	15.36	< 0.0001
`TV`	0.0475	0.0027	17.67	< 0.0001



Assessing the accuracy of the module: RSE

Quantify the extent to which the model fits the data

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$
$$= \sqrt{\frac{1}{n-2} \sum_i (y_i - \hat{y}_i)^2}$$

- estimate of the standard deviation of ε
- Avg amount that the response will deviate from the true regression line
- avg amount response will deviate from the true regression line

Assessing the accuracy of the module: R^2

R squared:

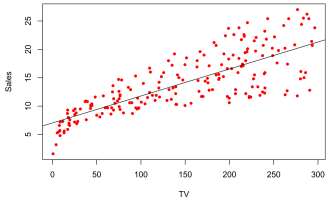
$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

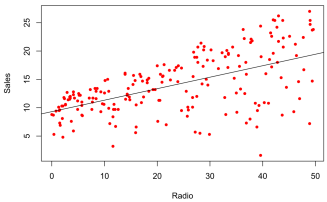
where total sum of squares is

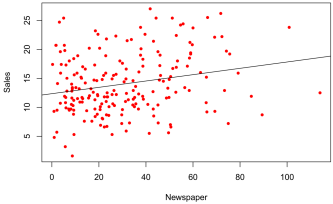
$$TSS = \sum_i (y_i - \overline{y})^2$$

- TSS is total variance in the response Y, variability before regression
- RSS amount of variability after the regression
- R^2 is proportion of variability in Y that can be explained using X
- Close to 1, large proportion of varaiability is explained by regression
- Close to 0, regression does not explain much of the variability in the response

Advertising example







$R^2 = 0.61$ $R^2 = 0.33$ $R^2 = 0.05$

Coding group work

Run the section titled "Assessing Coefficient Estimate Accuracy"

Point out that the homework uses the code slightly differently. statsmodels.formula.api vs statsmodels.api. You can use whatever you want on the homework.

Next time

Next time:

- Multi-linear regression.

CMSE381_S2026_Schedule : Sheet1

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression		HW #2 Due Sun 2/1		
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			
10	W	2/4	Logistic Regression	4.1, 4.2, 4.3.1-3		Q4	

Announcements

- Homework 1
  - Due Sun, Jan 25
- Homework 2
  - Due Sun, Feb 8